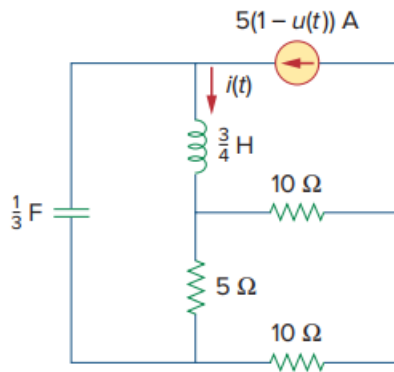


Problem

Refer to the circuit in Fig. 8.86. Calculate $i(t)$ for $t > 0$.

Figure 8.86:



Step-by-step solution

Step 1 of 15

Refer to Figure 8.86 in the text book.

At $t < 0$, the capacitor acts as open circuit and inductor acts as short circuit. The value of the current source is 5 A since the value of $1 - u(t)$ is 1 at $t < 0$.

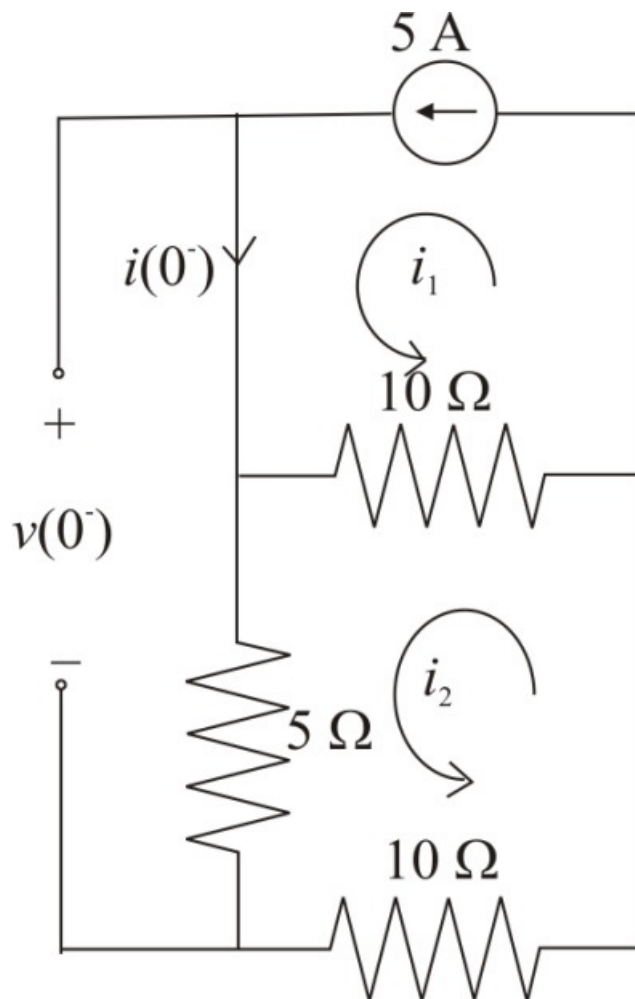


Figure 1

Step 2 of 15

The value of the current i_1 and hence $i(0^-)$ is 5 A .

Apply Kirchhoff's Voltage Law in loop-2.

$$10(i_2 - i_1) + 5i_2 + 10i = 0$$

$$10i_2 - 10i_1 + 15i_2 = 0$$

$$25i_2 - 10i_1 = 0$$

$$25i_2 = 10i_1$$

Substitute 5 A for i_1 .

$$25i_2 = 10(5)$$

$$25i_2 = 50$$

$$i_2 = \frac{50}{25}$$

$$i_2 = 2\text{ A}$$

Therefore, the value of the current i_2 is 2 A .

Step 3 of 15

Calculate the value of the voltage $v(0^-)$.

$$v(0^-) = 5i_2$$

Substitute 2 A for i_2 .

$$v(0^-) = 5(2)$$

$$= 10\text{ V}$$

Therefore, the value of the voltage $v(0^-)$ is 10 V .

Step 4 of 15

The value of the current source is 0 A since the value of $1-u(t)$ is 0 at $t > 0$. That is the circuit source can be treated as open circuit.

Draw the circuit diagram at $t > 0$.

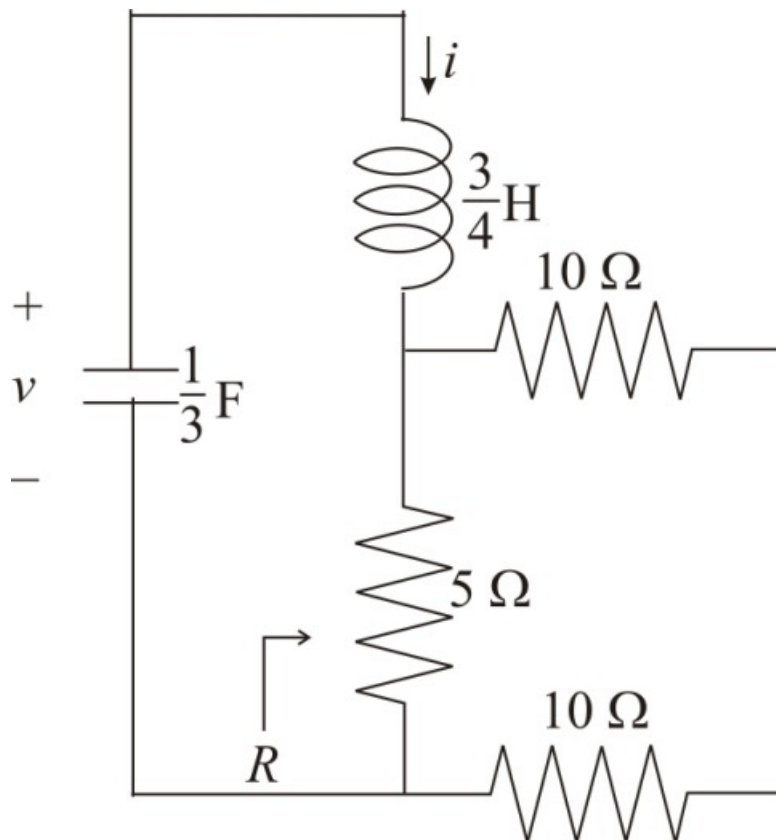


Figure 2

Step 5 of 15

The voltage across the capacitor and the current through the inductor will not change instantaneously. Hence, at $t > 0$ the value of the voltage across the capacitor $v(0^+)$ is 10 V and the value of the current $i(0^+)$ is 5 A .

Notice that the voltage across the capacitor and the current through the inductor are continuous. Therefore, the values at $t = 0$ are $v(0) = 10\text{ V}$ and $i(0) = 5\text{ A}$.

Step 6 of 15

Calculate the value of the resistance R .

$$\begin{aligned} R &= 5 \parallel (10 + 10) \\ &= 5 \parallel 20 \\ &= \frac{(20)(5)}{20 + 5} \\ &= \frac{100}{25} \end{aligned}$$

$$R = 4\ \Omega$$

Therefore, the value of the resistance R is $4\ \Omega$.

Step 7 of 15

Apply Kirchhoff's Voltage law in the loop.

$$\frac{3}{4} \frac{di}{dt} + Ri - v = 0$$

Substitute 4Ω for R .

$$\frac{3}{4} \frac{di}{dt} + 4i - v = 0$$

$$\frac{3}{4} \frac{di}{dt} = v - 4i$$

$$\frac{di}{dt} = \left(\frac{4}{3}\right)(v - 4i)$$

Calculate the value of $\frac{di}{dt}$ for $t = 0$.

$$\frac{di(0)}{dt} = \left(\frac{4}{3}\right)(v(0) - 4i(0))$$

Substitute 10 V for $v(0)$ and 5 A for $i(0)$.

$$\begin{aligned} \frac{di(0)}{dt} &= \left(\frac{4}{3}\right)(10 - 4(5)) \\ &= \left(\frac{4}{3}\right)(10 - 20) \\ &= \left(\frac{4}{3}\right)(-10) \\ &= -13.3333 \text{ V/s} \end{aligned}$$

Therefore, the value of $\frac{di(0)}{dt}$ is -13.3333 V/s .

Step 8 of 15

Apply Kirchhoff's Voltage Law in the loop.

$$\frac{3}{4} \frac{di}{dt} + Ri + \frac{1}{1/3} \int i \, dt = 0$$

Substitute 4Ω for R .

$$\frac{3}{4} \frac{di}{dt} + 4i + 3 \int i \, dt = 0$$

Apply derivative with respect to t .

$$\frac{3}{4} \frac{d^2i}{dt^2} + 4 \frac{di}{dt} + 3i = 0$$

Consider $\frac{d}{dt}$ to be s .

$$\frac{3}{4} s^2 i + 4si + 3i = 0$$

$$s^2 i + \frac{16}{3} si + 4i = 0$$

$$\left(s^2 + \frac{16}{3}s + 4\right)i = 0$$

Step 9 of 15

Consider the characteristic equation of the transient circuit.

$$s^2 + \frac{16}{3}s + 4 = 0$$

$$(s + 0.9028)(s + 4.4305) = 0$$

Therefore, the roots of the equation are $s_1 = -4.4305$ and $s_2 = -0.9028$.

The roots are negative and distinct. Therefore, the circuit exhibits over damped response.

Step 10 of 15

Draw the circuit diagram for $t = \infty$.

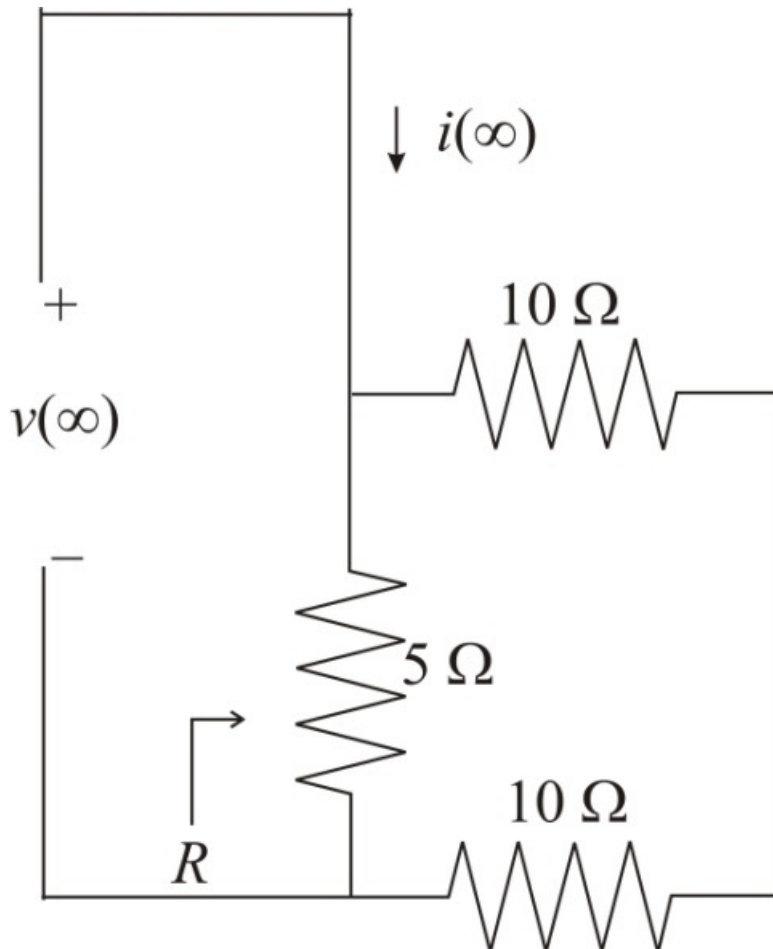


Figure 3

Therefore, the value of the current $i(\infty)$ is 0 A.

Step 11 of 15

Consider the expression of the current $i(t)$.

$$i(t) = i(\infty) + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Substitute 0 A for $i(\infty)$, -4.4305 for s_1 and -0.9028 for s_2 .

$$i(t) = 0 + A_1 e^{-4.4305t} + A_2 e^{-0.9028t}$$

$$i(t) = A_1 e^{-4.4305t} + A_2 e^{-0.9028t} \dots\dots (1)$$

Step 12 of 15

Recall equation (1).

$$i(t) = A_1 e^{-4.4305t} + A_2 e^{-0.9028t}$$

Substitute 0 for t .

$$\begin{aligned} i(0) &= A_1 e^{-4.4305(0)} + A_2 e^{-0.9028(0)} \\ &= A_1 e^0 + A_2 e^0 \\ &= A_1 (1) + A_2 (1) \\ &= A_1 + A_2 \end{aligned}$$

Substitute 5 A for $i(0)$.

$$5 = A_1 + A_2$$

$$A_2 = 5 - A_1 \dots\dots (2)$$

Step 13 of 15

Recall equation (1).

$$i(t) = A_1 e^{-4.4305t} + A_2 e^{-0.9028t}$$

Apply derivative with respect to t .

$$\begin{aligned} \frac{di(t)}{dt} &= A_1 \frac{d(e^{-4.4305t})}{dt} + A_2 \frac{d(e^{-0.9028t})}{dt} \\ &= -4.4305 A_1 e^{-4.4305t} - 0.9028 A_2 e^{-0.9028t} \end{aligned}$$

Substitute 0 for t and $5 - A_1$ for A_2 .

$$\begin{aligned} \frac{di(0)}{dt} &= -4.4305 A_1 e^{-4.4305(0)} - 0.9028 (5 - A_1) e^{-0.9028(0)} \\ &= -4.4305 A_1 e^0 - 0.9028 (5 - A_1) e^0 \\ &= -4.4305 A_1 (1) - (4.514 - 0.9028 A_1) (1) \\ &= -4.4305 A_1 - 4.514 + 0.9028 A_1 \end{aligned}$$

Substitute -13.3333 A/s for $\frac{di(0)}{dt}$

$$-13.3333 = -3.5277 A_1 - 4.514$$

$$3.5277 A_1 = 13.3333 - 4.514$$

$$3.5277 A_1 = 8.8193$$

$$A_1 = \frac{8.8193}{3.5277}$$

$$A_1 = 2.5$$

Therefore, the value of the constant A_1 is 2.5.

Step 14 of 15

Recall equation (2).

$$A_2 = 5 - A_1$$

Substitute 2.5 for A_1 .

$$\begin{aligned} A_2 &= 5 - 2.5 \\ &= 2.5 \end{aligned}$$

Therefore, the value of the constant A_2 is 2.5.

Step 15 of 15

Recall equation (1).

$$i(t) = A_1 e^{-4.4305t} + A_2 e^{-0.9028t}$$

Substitute 2.5 for A_1 and A_2 .

$$\begin{aligned} i(t) &= 2.5e^{-4.4305t} + 2.5e^{-0.9028t} \\ &= 2.5(e^{-4.4305t} + e^{-0.9028t}) \text{ A} \end{aligned}$$

Therefore, the value of the current $i(t)$ is $\boxed{2.5(e^{-4.4305t} + e^{-0.9028t}) \text{ A}}$.